SOP-IM-150-01: Start-up and production checks, Injection Moulding Machine IM-150 (SAMPLE, invented data)

This SOP is a test document. The machine and all values in it are invented and must not be used on a real machine.

Purpose: the steps an operator follows to start IM-150 at the beginning of a shift and the checks during production.
Applies to: operators and shift supervisors of the moulding shop. Revision B.

1. Before start-up

Wear heat-resistant gloves, safety glasses and safety shoes at the machine.
Test the safety gate interlock: open the gate and confirm that the mould cannot close. If the interlock test fails, stop the machine and tell the shift supervisor.
Check the hydraulic oil level in the sight glass and clean the hopper magnet.
Dry ABS granules in the hopper dryer at 80 degrees C for 2 hours before moulding.

2. Start-up

Switch on the barrel heaters and wait 30 minutes before the first shot, so that all zones reach the set temperature.
Open the cooling water valve. Cooling water at the inlet must be between 15 and 25 degrees C with a pressure of at least 3 bar.
Run 5 purge shots and discard them before the first production part.

3. Checks during production

The hydraulic oil temperature during production must be between 40 and 65 degrees C.
Write the oil temperature and the cooling water pressure in the shift log every 2 hours.
If alarm A-315 appears, stop production and call the maintenance technician.

4. Stopping

If the machine will stand still for more than 30 minutes, lower the barrel temperature to the standby value of 150 degrees C.
At the end of the shift, retract the injection unit and leave the mould slightly open.